

Regular Expression

Atomic Patterns.

$a \in \Sigma$

ϵ

ϕ

Compound Patterns

$+$

$\cdot$

$*$

Patterns constructed from the above syntax are called Regular Expressions.

Precedence: $* > \cdot > +$ (left to right)

Ex: $\alpha + \beta \gamma \rightarrow \alpha + (\beta \gamma)$ and not $(\alpha + \beta) \gamma$

$\alpha \beta^* \rightarrow \alpha (\beta^*)$ and not $(\alpha \beta)^*$

Note: Use parenthesis.

Theorem. Let $A \subseteq \Sigma^*$. The following statements are equivalent.

1. A is regular. $\exists$ a finite automaton M s.t. $L(M) = A$.
2. $A = L(\alpha)$ for some pattern α .
3. $A = L(\alpha)$ for some regular expression α .

Proof. $3 \Rightarrow 2$ is trivial.

$2 \Rightarrow 1$

$1 \Rightarrow 3$: To convert an automata M to an equivalent regular expression

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Proof $2 \Rightarrow 1$.

Atomic pattern

$a \in \Sigma$

ϵ

$\emptyset$

$\#$
 $@$ } Redundant

Compound pattern

$\beta + \gamma$

$\beta \cap \gamma$ •

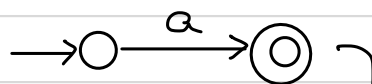
$\beta \gamma$

β^*

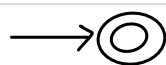
β^+ •

$\neg \beta$

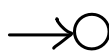
$a \in \Sigma \quad L(a) = \{a\}$



$\epsilon: L(\epsilon) = \{\epsilon\}$



$\emptyset: L(\emptyset) = \emptyset$



} Regular

$\beta + \gamma$: $L(\beta + \gamma) = L(\beta) \cup L(\gamma)$ (Definition)

By IH $L(\beta)$ is regular and $L(\gamma)$ is regular.

Regular sets are closed under union
So $L(\beta + \gamma)$ is regular.

$\beta \cap \gamma$: $L(\beta \cap \gamma) = L(\beta) \cap L(\gamma)$ (Definition)

By IH $L(\beta)$ is regular and $L(\gamma)$ is regular

Regular sets are closed under intersection
So $L(\beta \cap \gamma)$ is regular

$\beta \gamma$: $L(\beta \gamma) = L(\beta) L(\gamma)$ (Definition)

By IH, $L(\beta)$ and $L(\gamma)$ are regular.

Regular sets are closed under concatenation
So $L(\beta \gamma)$ is regular.

β^* : $L(\beta^*) = L(\beta)^*$ (Defn.)

By IH, $L(\beta)$ is regular; Regular sets are closed under $*$
So $L(\beta^*)$ is regular.

$\neg \beta$: $L(\neg \beta) = \overline{L(\beta)}$ (Defn.)

By IH $L(\beta)$ is regular.

Regular sets are closed under complementation
So $L(\neg \beta)$ is regular.

Theorem. Let $A \subseteq \Sigma^*$. The following statements are equivalent.

1. A is regular. $\exists$ a finite automaton M s.t. $L(M) = A$.
2. $A = L(\alpha)$ for some pattern α .
3. $A = L(\alpha)$ for some regular expression α .

$1 \Rightarrow 3$ Given a finite state automaton M , we can construct a regular expression α s.t. $L(\alpha) = L(M)$.

Proof:

Let $M = (Q, \Sigma, \Delta, S, F)$ be an NFA without ϵ transitions

For all $Y \subseteq Q$ and $u, v \in Q$ we construct a regular expression α_{uv}^Y

α_{uv}^Y - the set of all strings x such that

there is a path from state u to state v in M

labelled x [Formally, $v \in \hat{\Delta}(\{u\}, x)$]

and

all states along the path with possible exception of u and v lie in Y .

Induction on size of γ .

Base Case : $\gamma = \emptyset$.

$$a_1, a_2, \dots, a_k \in \Sigma \text{ s.t. } v \in \Delta(u, a_i)$$

Case 1. $u \neq v$

$$\alpha_{uv}^{\emptyset} = \begin{cases} a_1 a_2 \dots a_k & \text{if } k \geq 1. \\ \emptyset & \text{if } k = 0 \end{cases}$$

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Case 2. $u = v$

$$\alpha_{uv}^{\emptyset} = \begin{cases} a_1 + a_2 + \dots + a_k + \epsilon & \text{if } k \geq 1 \\ \epsilon & \text{if } k = 0 \end{cases}$$

Induction on size of γ

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Induction Step.

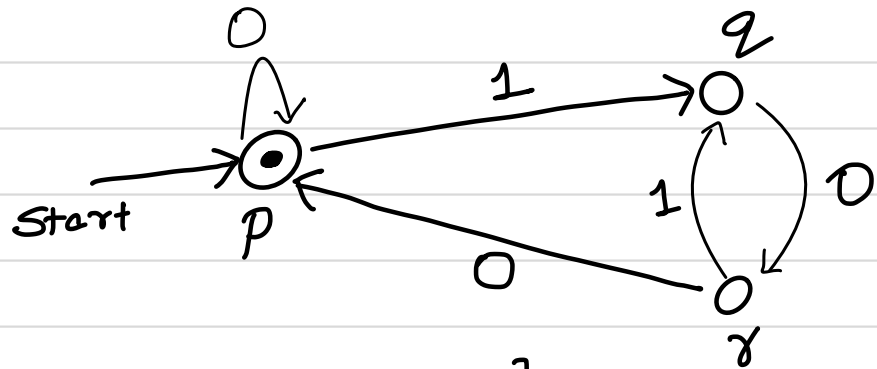
$$\alpha_{uv}^{\gamma} = \alpha_{uv}^{\gamma - \{q\}} + \alpha_{uq}^{\gamma - \{q\}} (\alpha_{qq}^{\gamma - \{q\}})^* \alpha_{qv}^{\gamma - \{q\}}$$

Choose an arbitrary state $q \in Y$

α - Sum of all expressions of the form.

$$\alpha_{sf}^Q \text{ where } s \in S \text{ and } f \in F.$$

Example:

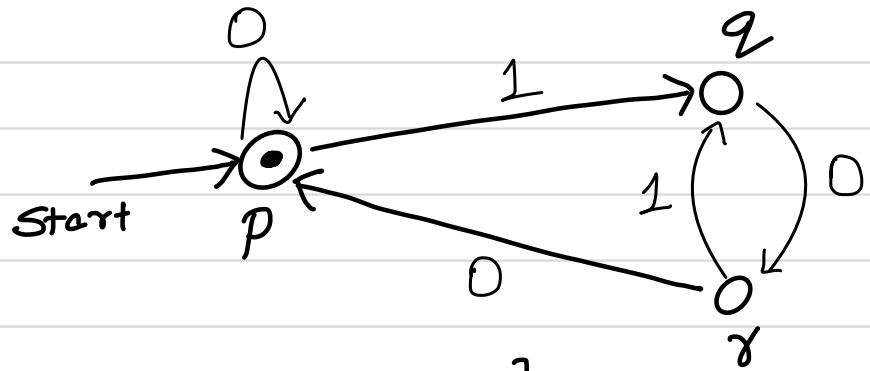


Regular expression: $\alpha_{pp}^{\{p,q,r\}}$

Removing state q

$$\alpha_{pp}^{\{p,q,r\}} = \alpha_{pp}^{\{p,r\}} + \alpha_{pq}^{\{p,r\}} (\alpha_{qq}^{\{p,r\}})^* \alpha_{rp}^{\{p,r\}},$$

Example:



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Removing state q

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$$\alpha_{pp}^{\{p,r\}} = 0^*$$

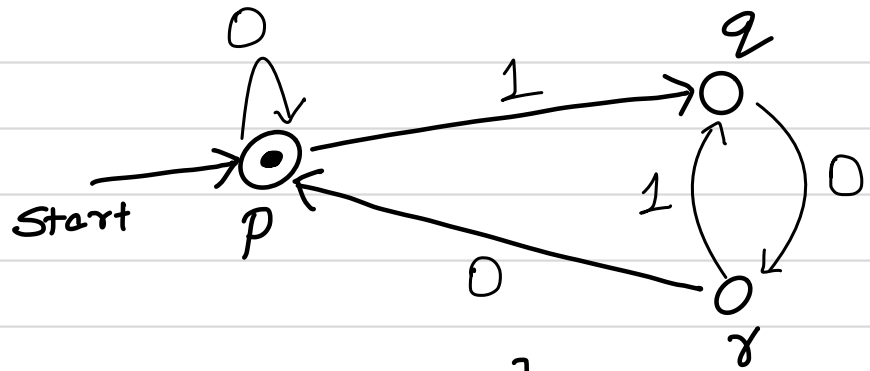
$$\alpha_{pq}^{\{p,r\}} = 0^* 1$$

$$\alpha_{qq}^{\{p,r\}} = \epsilon + 01 + 000^*1$$

$$= \epsilon + 0(\epsilon + 000^*)1 = \epsilon + 000^*1$$

$$\alpha_{qp}^{\{p,r\}} = 000^*$$

Example:



Regular expression: $\alpha_{pp}^{\{p,q,r\}}$

Removing state q

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